

ASSIGNMENT #11 November 15, 2022

COORDINATOR: SASHA

SECTION 7.4

Exercise (7.4.7 — Stefano Fochesatto). Let R be a commutative ring with 1. Prove that the principal ideal generated by x in the polynomial ring $R[x]$ is a prime ideal if and only if R is an integral domain. Prove that (x) is a maximal ideal if and only if R is a field.

Proof. Let R be a commutative ring with 1. Consider the homomorphism $\varphi : R[x] \rightarrow R$ defined by $\varphi(a_0 + a_1x + a_2x^2 + \dots) = a_0$. Let $p(x), q(x) \in R[x]$ such that $\varphi(p(x)) = a_0$ and $\varphi(q(x)) = b_0$. Note that by polynomial multiplication $\varphi(p(x)q(x)) = a_0b_0 = \varphi(p(x))\varphi(q(x))$ and clearly φ is a surjection. Also note that every $p(x) \in \ker(\varphi)$ must have zero constant term and therefore is comprised of products and sums of x so $\ker(\varphi) = (x)$.

By The First Isomorphism Theorem for Rings we find that $R[x]/(x) \cong R$. By Proposition 13, supposing (x) is a prime ideal in $R[x]$ we know that $R[x]/(x)$ is an integral domain, thus R is an integral domain. Since Proposition 13 is an if and only if statement if we suppose R is an integral domain then we get that $R/(x)$ is an integral domain and therefore (x) is a prime ideal.

Supposing (x) is a maximal ideal we get that $R/(x)$ is a field by Proposition 12 and therefore R is a field. Again since Proposition 12 is an if and only if statement, supposing R is a field we know that $R/(x)$ is a field and therefore (x) must be a maximal ideal.

□

Exercise (7.4.41 — Stefano Fochesatto). A proper ideal Q of the commutative ring R is called primary if whenever $ab \in Q$ and $a \notin Q$ then $b^n \in Q$ for some positive integer n . Establish the following facts about primary ideals,

- (a) The primary ideals of $\mathbb{Z}$ are 0 and (p^n) , where p is a prime and n is a positive integer.

Proof. Clearly 0 is a primary ideal of $\mathbb{Z}$. Let $ab \in (p^n)$ then either $p \mid a$ or $p \mid b$ which implies either $a^n \in (p^n)$ or $b^n \in (p^n)$. Suppose for the sake of contradiction (i) is a prime ideal where i is not prime or prime power. Let $i = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_n^{\alpha_n}$ where p_k are prime and note that by definition $(p_1^{\alpha_1} p_2^{\alpha_2} \dots p_{n-1}^{\alpha_{n-1}})(p_n^{\alpha_n}) = i \in (i)$ however clearly neither $p_n^{\alpha_n} \in (i)$ or $(p_1^{\alpha_1} p_2^{\alpha_2} \dots p_{n-1}^{\alpha_{n-1}}) \in (i)$ so (i) is not primary. $\square$

- (b) Every prime ideal of R is a primary ideal.

Proof. Suppose P is a prime ideal. By definition if $ab \in P$ then either $a \in P$ or $b \in P$. Without loss of generality suppose $a \notin P$ and $b \in P$, then clearly $b^n \in P$ and we find that P is primary. $\square$

- (c) An ideal Q of R is primary if and only if every zero divisor in R/Q is a nilpotent element of R/Q .

Proof. Suppose Q , an ideal of R is primary. Let $(i + Q) \in R/Q$ be a zero divisor, so $(j + Q) \in R/Q$ such that $(i + Q)(j + Q) = (ij + Q) = Q$ where $(i + Q), (j + Q) \neq Q$. Therefore $ij \in Q$ and $j \notin Q$ since Q is primary, there exists some $i^n \in Q$ so we get $(i + Q)^n = (i^n + Q) = Q$. Thus $(i + Q)$ is nilpotent in R/Q .

Suppose every zero divisor in R/Q is a nilpotent element of R/Q . Let $ab \in Q$ and $a \notin Q$ note that $(ab + Q) = (a + Q)(b + Q) = (a + Q)(Q) = Q$. Thus $(b + Q)$ is a zero divisor. Then there exists some $(b + Q)^n = (b^n + Q) = Q$ so $b^n \in Q$. Therefore Q is primary. $\square$

- (d) If Q is a primary ideal then $\text{rad}(Q)$ is a prime ideal.

Proof. Suppose Q is a primary ideal. Let $ab \in \text{rad}(Q)$. By definition $\text{rad}(Q)$ there exists some n such that $(ab)^n \in Q$. Since R is commutative we know that $(a^n)(b^n) \in$

Q , without loss of generality suppose $(a^n) \notin Q$, since Q is primary then there exists some $b^{mn} \in Q$. Therefore $b \in \text{rad}(Q)$. $\square$